

Fano threefolds and fourfolds admitting non-isomorphic endomorphisms

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1. Background & Motivation
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3. Main Result & Proofs
4. Open Questions

This talk is based on the following joint works.

[MZZ20]

Non-isomorphic endomorphisms of Fano threefolds (arXiv:2008.10295),
Sheng Meng, De-Qi Zhang and Guolei Zhong.

[JZ20]

Amplified endomorphisms of Fano fourfolds (arXiv:2011.11239),
Jia Jia and Guolei Zhong.

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- (curve case): Hurwitz formula $\Rightarrow X$ is either rational or elliptic.
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 - (surface case): Such (possibly singular) X has been fully studied by Nakayama ([Nak08], [Nak20, RIMS-1920], [Nak20, RIMS-1923]).
 - (higher dimension): There are three (equivariant) typical fibrations to help us study such X : **Albanese map**, **Kodaira fibration**, and **maximal rationally connected fibration** (MRC).
(Reduce to the study of the base and general fibres.)

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(Iitaka fibration, Albanese map and MRC fibration)
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- A normal projective variety X of dimension n is a **toric variety** if X contains a big torus $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on the whole variety. We say $(X, X \setminus T)$ is a **toric pair**.
components of $X \setminus T = \dim X + \rho(X)$ when X is $\mathbb{Q}$ -factorial.

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Question (Fakhruddin, 2003 (resp. Meng-Z., 2020))

Let X be a smooth projective rationally connected variety admitting a polarized (resp. int-amplified) endomorphism. Will X be a toric variety?

Early results

Theorem (Nakayama, 2002 (Sato's conjecture))

A smooth rational surface is toric if and only if it admits a non-isomorphic surjective endomorphism. (Rmk: ✓ in dimension two)

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Theorem (Meng-Zhang, 2019 (resp. Meng-Z., 2020))

Let X be a rationally connected smooth projective variety with D a reduced divisor. Then (X, D) is a toric pair if and only if X admits a polarized (resp. int-amplified) endomorphism f such that $f|_{X \setminus D}$ is étale. (Rmk: "TIR conditions", generalize [Hwang-Nakayama08] $\rho(X) = 1$ case)

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Analysing conic bundle structures is a main topic in this project.

Facts on the conic bundle structures

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- When Y is a surface and $\Delta_\tau \neq \emptyset$, if y is a smooth point of Δ_τ , then X_y is a reducible conic in $\mathbb{P}^2$; if y is a double point of Δ_τ , then X_y is a double line in $\mathbb{P}^2$.

Proofs for the main results (Threefold) - - Conic bundle

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- If $\rho(X) \geq 3$, then X admits a (not necessarily elementary) conic bundle structure (Mori-Mukai 1983 + dynamical assumption).

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$\tau \xRightarrow{(i)}$ conic bundle $\xRightarrow{(ii)}$ (smooth) $\mathbb{P}^1$ -bundle, i.e., $\Delta = \emptyset$

$\xRightarrow{(iii)}$ algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$

$\xRightarrow{(iv)}$ splitting $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{O} \oplus \mathcal{L})$

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(i) Fano contr. of rel dim one $\implies$ (elementary) conic bundle

Mori's (1982) motivating result (for threefold). In general, Ando (1985) showed that if the Fano contr. (from a smooth projective variety) is equi-dimensional of rel dim one, then it is a conic bundle.

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Let $\tau : X \rightarrow Y$ be a (smooth) $\mathbb{P}^1$ -bundle over a smooth projective rational variety Y . Then $X = \mathbb{P}_Y(\mathcal{E})$ for some locally free rank-two sheaf $\mathcal{E}$ over Y . (without using dynamics)

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(iv) Motivation to find a splitting conic bundle?

If $X \rightarrow Y$ is a splitting $\mathbb{P}^1$ -bundle and (Y, Δ_Y) is a toric pair, then X is toric and $(X, S_0 + S_1 + \tau^{-1}(\Delta_Y))$ is a toric pair for any two disjoint (cross) sections S_0, S_1 of τ .

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- The second contr. $\pi : X \rightarrow Z$ is Fano with $Z \cong Y$.
- **Prove the case $Y \cong \mathbb{P}^2$:** Note that $S_i := \tau^{-1}(C_i)$ are f^{-1} -invariant and $\ell := S_1 \cap S_2$ cannot be contracted by π . Since ℓ is f^{-1} -invariant, $D := \pi^{-1}(\pi(\ell))$ is an f^{-1} -invariant prime divisor. Then we found three f^{-1} -invariant divisors S_1 , S_2 and D sharing a common intersection ℓ . So the pair $(X, S_1 + S_2 + D)$ is not lc at ℓ , a contradiction to (Broustet-Höring 2014).

(iv) splittingness of algebraic $\mathbb{P}^1$ -bundle (Elementary case)

(Amerik, 2003) Let $\tau : X = \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ be an algebraic $\mathbb{P}^1$ -bundle over a smooth projective rational variety. Suppose $\text{rank } \mathcal{E} = 2$ and X admits a non-isomorphic endomorphism f . Then τ splits if either

- $\deg f > \deg f|_Y$ (this is the case when f is IAMP) and $\mathcal{E}$ is H -semistable for some ample divisor H ; or
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(iv) splittingness of algebraic $\mathbb{P}^1$ -bundle (Birational dominance case)

Let X be a smooth Fano threefold admitting an IAMP endomorphism f . Suppose that X admits a conic bundle $\tau : X \rightarrow Y$, which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ with τ_0 being a (smooth) $\mathbb{P}^1$ -bundle. Then τ_0 is a splitting $\mathbb{P}^1$ -bundle. (Here, W is not necessarily Fano!)

Proofs for the main results (Threefold) - - Conic bundle

Equivariant minimal model program for conic bundle structures

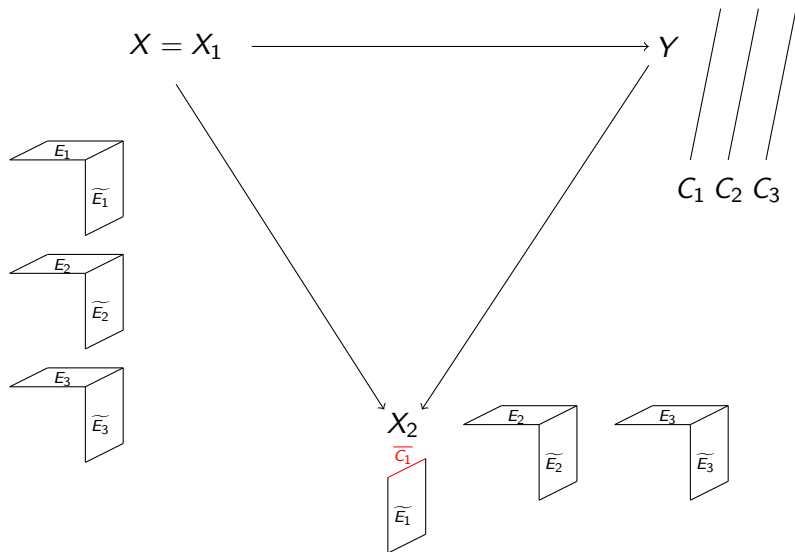
Let X be a smooth Fano threefold with a conic bundle $\tau : X \rightarrow Y$, and $f : X \rightarrow X$ a non-isomorphic surjective endomorphism. Then there exists an f -equivariant (after iteration) MMP

$$X = X_1 \rightarrow \cdots \rightarrow X_i \rightarrow \cdots \rightarrow X_{r+1} \rightarrow Y$$

such that the following hold.

- ① $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth Fano threefold.
- ② $\tau_{r+1} : X_{r+1} \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y .
- ③ $\tau_i : X_i \rightarrow Y$ is a conic bundle with $(f|_Y)^{-1}$ -invariant discriminant $\Delta_{\tau_i} = C_i \cup \cdots \cup C_r$ a disjoint union of $r + 1 - i$ smooth curves on Y .
- ④ $\pi : X \rightarrow X_{r+1}$ is the blowup of X_{r+1} along r disjoint union of $(f|_{X_{r+1}})^{-1}$ -invariant smooth curves $\bigcup_{i=1}^r \overline{C_i}$ with $\tau_{r+1}(\overline{C_i}) = C_i$.

Proofs for the main results (Threefold) - - Conic bundle



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Theorem (Reduce to polarized case (skip the proof))

Let f be a non-isomorphic surjective endomorphism of a smooth Fano threefold X . Then either f is polarized after iteration or $X \cong Y \times \mathbb{P}^1$ for a del Pezzo surface Y .

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Let X be a smooth Fano threefold with $\rho(X) \geq 3$. Suppose that X admits an IAMP endomorphism f . Then X is toric.

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Idea for the proof

The key point for the proof of the above theorem is to show that X is a (toric) blow-up of a (not necessarily Fano) splitting $\mathbb{P}^1$ -bundle over a toric surface along disjoint curves which are intersections of some f^{-1} -invariant prime divisors.

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Since $\rho(X) \geq 3$, we have a conic bundle $\tau : X \rightarrow Y$. Then it factors as $X = X_1 \rightarrow \cdots \rightarrow X_{r+1} = \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$.

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- (Mori-Mukai, 1983) $\tau^{-1}(C_i) = E_i \cup F_i$, and E_i, F_i are $\mathbb{P}^1$ -bundles over C_i (One is exceptional divisor and another is the proper transform of $\tau_{r+1}^{-1}(C_i)$).

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- There is an (f -equivariant) MMP $X \rightarrow X'$ over Y contracting all E_i such that $\tau' : X' \rightarrow Y$ is a splitting $\mathbb{P}^1$ -bundle by **Splittingness of algebraic $\mathbb{P}^1$ -bundle (Birational dominance case)**.

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- π' can be viewed as the blow-up of X' along r smooth curves $C'_i := F'_i \cap S'$ where $F'_i := \tau'^{-1}(C_i)$ and $S' := \pi'(S)$.

Proofs for the main results (Threefold)

Remark (a question in the above proof)

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Reason (Theorem, Meng-Zhang-Z., 2020):

Let $\tau : X \rightarrow Y$ be a splitting $\mathbb{P}^1$ -bundle over a smooth projective rational surface Y . Suppose that X admits an IAMP endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union Σ of all the f^{-1} -periodic prime divisors.

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The end of the proof

π' is the composition of toric blow-ups of the intersection of two prime divisors in the log smooth toric boundary starting from the (log smooth) toric pair (X', Δ') . We are done. ✓

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What's the connection and the differences

- The appearance of small contr. makes things much more complicated when we deal with fourfold.
- How to deal with the case of Fano fourfolds of Picard number one admitting non-isomorphic endomorphisms? Will it be $\mathbb{P}^4$?
- The Fano contr. from a Fano fourfold may not be equi-dimensional, even if it is of relative dimension one ($\ncong$ a conic bundle).
- Can we eliminate the special fibre using the dynamical condition?

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- Can we eliminate the special fibre using the dynamical condition?
- We start from a (not necessarily elementary) conic bundle τ .
- We show that there is an equivariant MMP for τ .
- We also show that an elementary conic bundle under the dynamic assumption is smooth and further a splitting $\mathbb{P}^1$ -bundle.

Proofs for the main results (Fourfold)

Equivariant minimal model for conic bundles (fourfold)

Let X be a smooth Fano fourfold and $\tau: X \rightarrow Y$ a conic bundle. Suppose X admits an IAMP endomorphism f . Then, after iteration, there exists an f -equivariant minimal model program

$$X = X_1 \rightarrow \cdots \rightarrow X_{r+1} = W \rightarrow Y$$

such that the following assertions hold.

- $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth manifold.
- $\tau_0: W \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle over a smooth Fano threefold.
- There are r disjoint prime divisors $D_1, \dots, D_r$ on Y and r pairs of prime divisors $E_i, \tilde{E}_i$ on X such that $\tau^* D_i = E_i + \tilde{E}_i$ and $D_i \cap D_j = \emptyset$.
- τ_0 is smooth and $\Delta_\tau = \bigsqcup_{i=1}^r D_i$ with each D_i an $(f|_Y)^{-1}$ -invariant smooth rational connected component of Δ_τ . Moreover, τ has reduced fibres over $\bigcup_{i=1}^r D_i$.

Proofs for the main results (Fourfold)

Equivariant minimal model for conic bundles (continued)

- The composition $X_i \rightarrow Y$ is a conic bundle with the discriminant $D_i \cup \dots \cup D_r$.
- The composition $\pi: X \rightarrow W$ is the blow-up of W along r disjoint union of $(f|_W)^{-1}$ -invariant smooth rational surfaces $\bigcup_{i=1}^r \overline{D}_i$ with $\tau_0|_{\overline{D}_i}: \overline{D}_i \cong D_i$.

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Splittingness of elementary conic bundle (fourfold)

Let X be a smooth Fano fourfold admitting an IAMP endomorphism f . Suppose $\tau: X \rightarrow Y$ is a conic bundle which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$. If one of the following holds, then τ_0 splits.

- $\rho(X/W) \neq 1$, i.e., either $\tau = \tau_0$ is elementary or $\rho(X/W) \geq 2$;
- $Y \cong \mathbb{P}^3$ or the blow-up of a line on $\mathbb{P}^3$;
- Y is a $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, $\mathbb{F}_0$ or $\mathbb{F}_1$.

Proofs for the main results (Fourfold)

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Furthermore, if τ_0 is not a splitting $\mathbb{P}^1$ -bundle, then there exists another conic bundle $\widehat{\tau}: X \rightarrow \widehat{W} \rightarrow \widehat{Y}$ with $\widehat{\tau}_0: \widehat{W} \rightarrow \widehat{Y}$ being elementary such that $\rho(X) - \rho(\widehat{W}) = 2$. In particular, $\widehat{W}$ is a splitting $\mathbb{P}^1$ -bundle over Y .

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A main difference when compared with threefold case

- (In fourfold case) we can't show the splittingness of τ_0 when π is a single blow-up. However, the splittingness is a consequence of (Jia-Z., 2020) since the toric action descends along a fibration.
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With a similar argument as in threefold case, we have:

Let X be a smooth Fano fourfold admitting a conic bundle. Suppose that X admits an IAMP endomorphism. Then X is toric. ✓

Conjecture

Let $f : X \rightarrow X$ be a non-isomorphic surjective endomorphism on the projective space $X = \mathbb{P}^n$. Let Z be a totally periodic closed subvariety, i.e., $f^{-1}(Z) = Z$ after iteration. Will Z be a linear subspace?

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The above conjecture has been confirmed in the following cases:

- $X = \mathbb{P}^2$ (Fornaess-Sibony 1994, Gurjar 2003).
- $X = \mathbb{P}^3$ and Z is an f^{-1} -periodic prime divisor (Nakayama-Zhang 2010, H\"oring 2017) .
- Z is a smooth hypersurface of X (Cerveau-Lins Neto 2000, Beauville 2001, Gurjar 2003).

Open questions

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A positive answer to the above conjecture in $\dim X = 4$ will, to a great extent, help us solve our toric problem completely in $\dim X = 4$.

Question (toric boundary conjecture)

Let $f : X \rightarrow X$ be an IAMP (or polarized) endomorphism of a rationally connected smooth projective variety X . Let Σ be the union of f^{-1} -periodic divisors. Is X toric, and if so, is there a toric pair (X, Δ) such that $\Sigma \subseteq \Delta$?

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Remark

The above question has a positive answer in the following cases:

- ① $\dim X = 2$ (Meng-Zhang-Z. 2020).
- ② X is a smooth Fano threefold (Meng-Zhang-Z. 2020, Jia-Z. 2020).

f^{-1} -periodic subvarieties of higher codimension will be much harder. For example, $\mathbb{P}^2$ has at most three f^{-1} -invariant points?

Thank You!